CSC1058 USER INTERFACE DESIGN & IMPLEMENTATION

MODULE COORDINATOR & LECTURER
Hyowon Lee

TEACHING ASSISTANT
Haoyan Zhang

	Tuesday (3 – 4pm)	Wednesday (11am - 12pm)
Week 1 (7 Sep)	Introduction/ orientation	HCI background
Week 2 (14 Sep)	Design guidelines	Design guidelines
Week 3 (21 Sep)	Design guidelines	Design guidelines
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Week 5 (5 Oct)	Guidelines recap	Guidelines recap
Week 6 (12 Oct)	Preventing negative impact	Usability evaluation
Week 7 (19 Oct)	Usability evaluation	Usability evaluation
Week 8 (26 Oct)	Usability evaluation	Usability evaluation
Week 9 (2 Nov)	Usability evaluation	Usability evaluation
Week 10 (9 Nov)	Design principles summary	Visualisation summary
Week 11 (16 Nov)	Project discussion	Project feedback
Week 12 (23 Nov)	Summary/ reflection	Summary/ reflection

DCU

1

Guideline 3: Speak the user's language

Instruction, message, icons, style… from the target user's point of view

DCU

2

Guid

"We have sold you 2 items from the shelf"

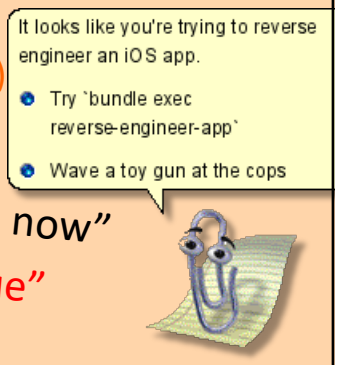
➡ "You have bought 2 items from the shelf"
(user's point of view)

"Action preference #327 initiated"

➡ "Background setting ready"
(no system-oriented information)

"I completed phase 1, you can continue now"

➡ "Phase 1 complete. Please continue"
(non-anthropomorphic dialogue)



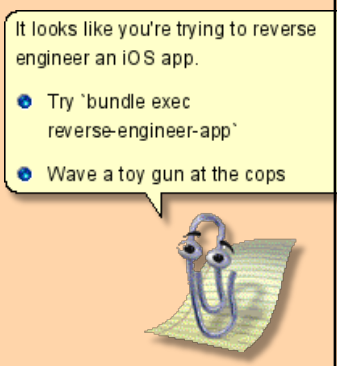
3

Guid

"I will begin the lesson when you press RETURN."

"You can begin the lesson by pressing RETURN."

"To begin the lesson, press RETURN."



Read more on non-anthropomorphic design... Shneiderman book



Above example from Shneiderman book

4

Anthropomorphism: the attribution of human traits, emotions, or intentions to non-human entities

…e lesson when you press RETURN."

…the lesson by pressing RETURN."

…sson, press RETURN."

It looks like you're trying to reverse engineer an iOS app.

- Try `bundle exec reverse-engineer-app`
- Wave a toy gun at the cops

Read more on non-anthropomorphic design... Shneiderman book



Above example from Shneiderman book

5

Guid…

"I will begin the lesson when you press RETURN."

"Y… …RN."

Non-anthropomorphic design guidelines

- Be cautious in presenting computers as people, either with synthesised or cartoon characters.
- Design comprehensible, predictable, and user-controlled interfaces.
- Use appropriate humans for audio or video introductions or guides.
- Use cartoon characters in games or children's software, but avoid them elsewhere.
- Provide user-centred overviews for orientation and closure.
- Do not use "I" when the computer responds to human actions.
- Use "you" to guide users, or just state facts.

…you're trying to reverse …iOS app.

…dle exec …engineer-app`

…oy gun at the cops

…neiderman …ook

6

Guideline 3:
Speak the user's language

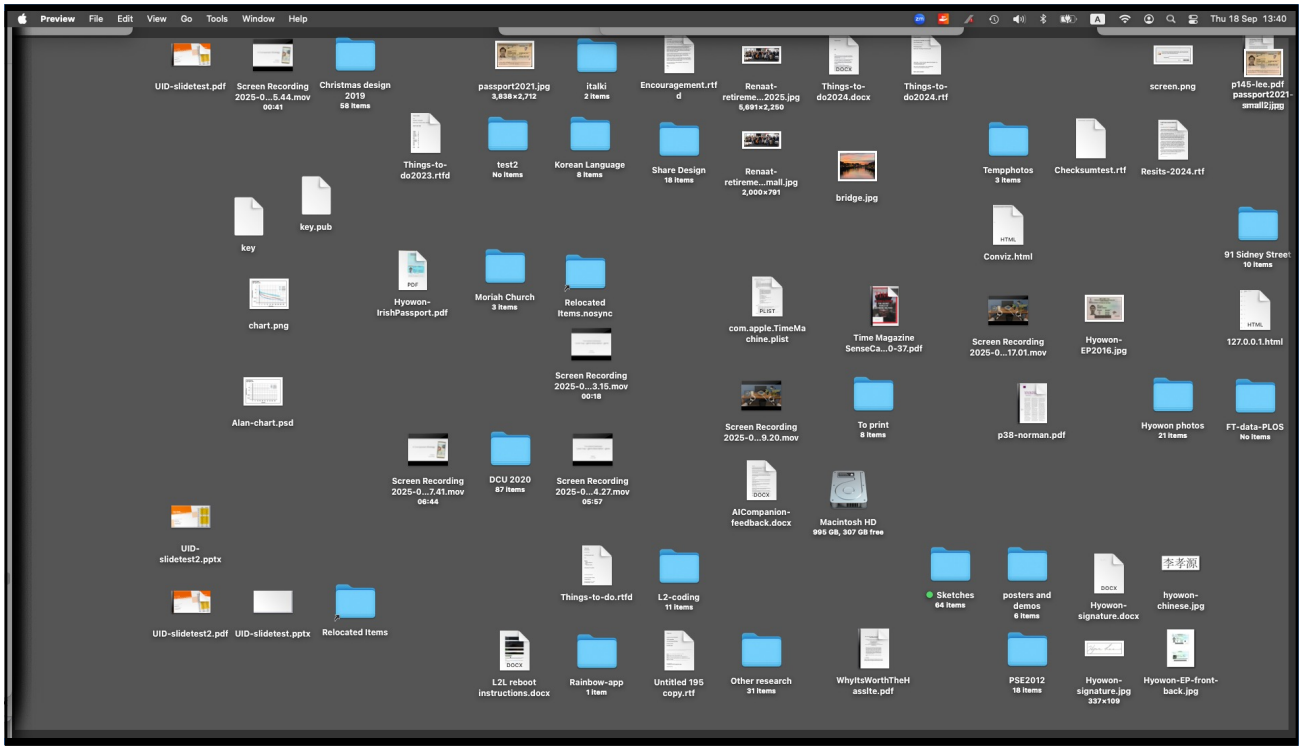
Instruction, messa
the target user's p

Metaphor

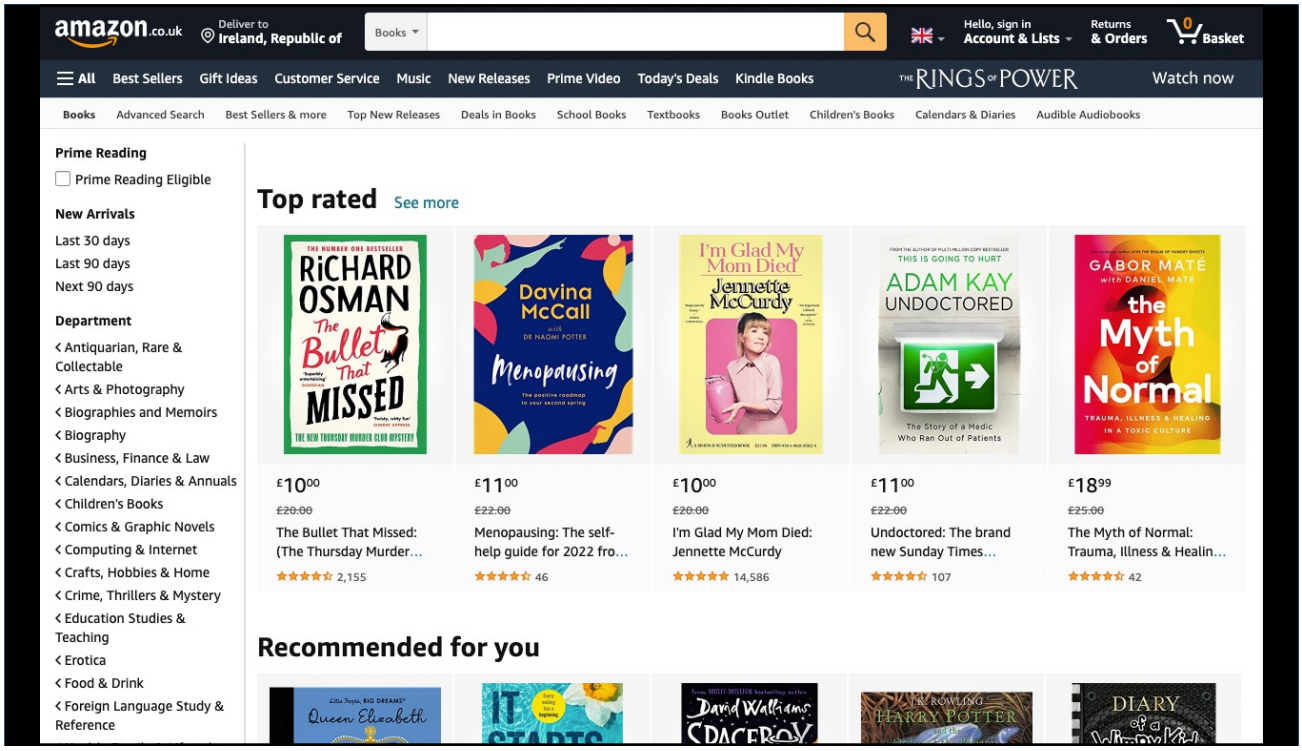
Metaphor: a word or phrase for one thing that is used to refer to another thing in order to show or suggest that they are similar

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7



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Guideline 3: Speak the user's language

Instruction, message, icons, style… from the target user's point of view

Metaphor

… leveraging user's knowledge on real world

… where metaphor breaks!

5 Usability Criteria

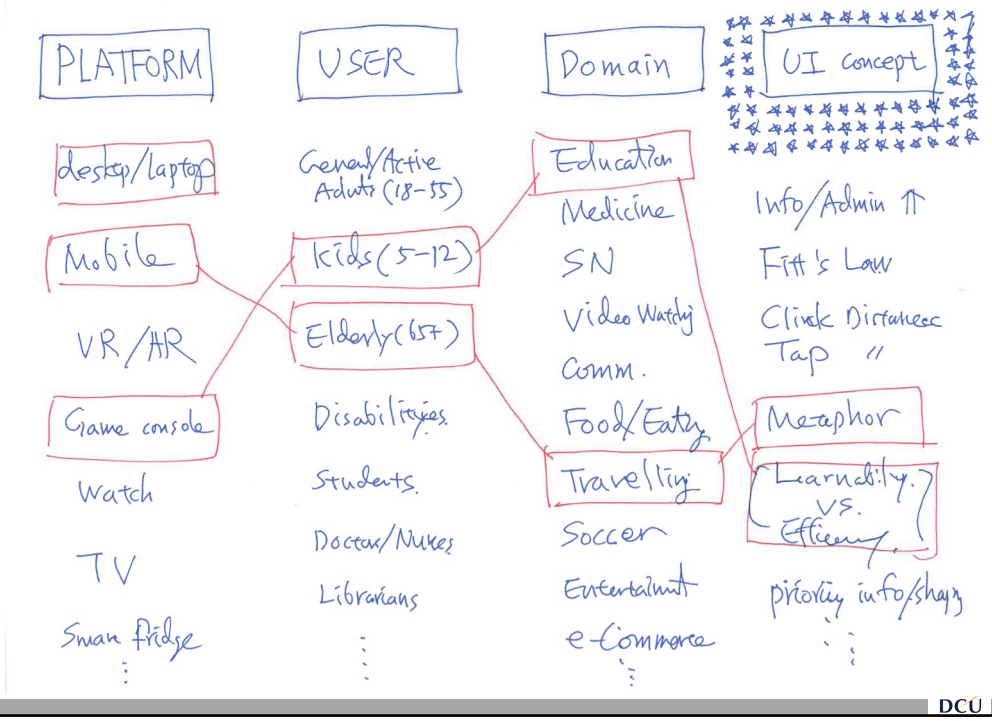
- Efficiency
- Learnability
- Memorability
- Error rate
- Satisfaction

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10

Framing the Project

11



12

Guideline 4: Provide informative feedback

- UI should provide clues on user's action
- While waiting?

Computation power, graphics processing, network bandwidth…

User's expectation



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Guideline 4: Provide informative feedback

- UI should provide clues on user's action
- While waiting?
- Direct Manipulation

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14

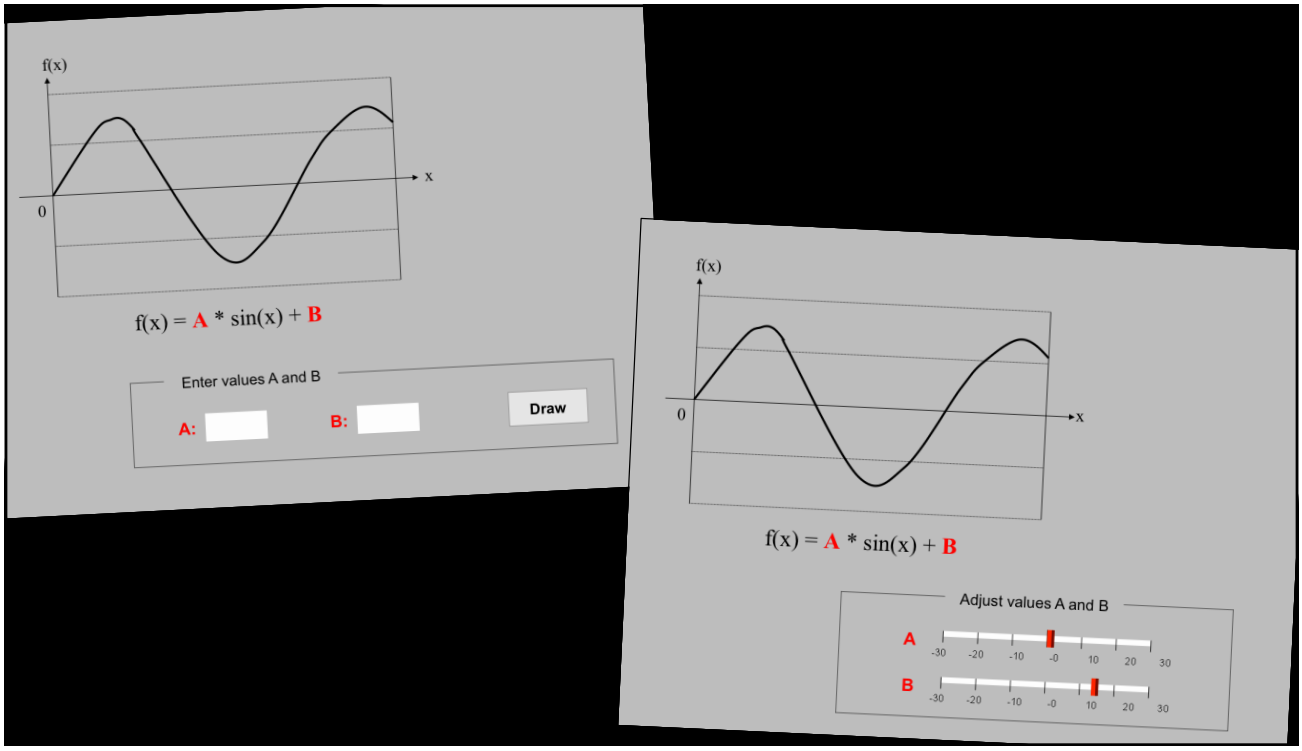
Direct Manipulation

- Immediate and continuous feedback
- Incremental and reversable user action
- No user-driven error

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DCU

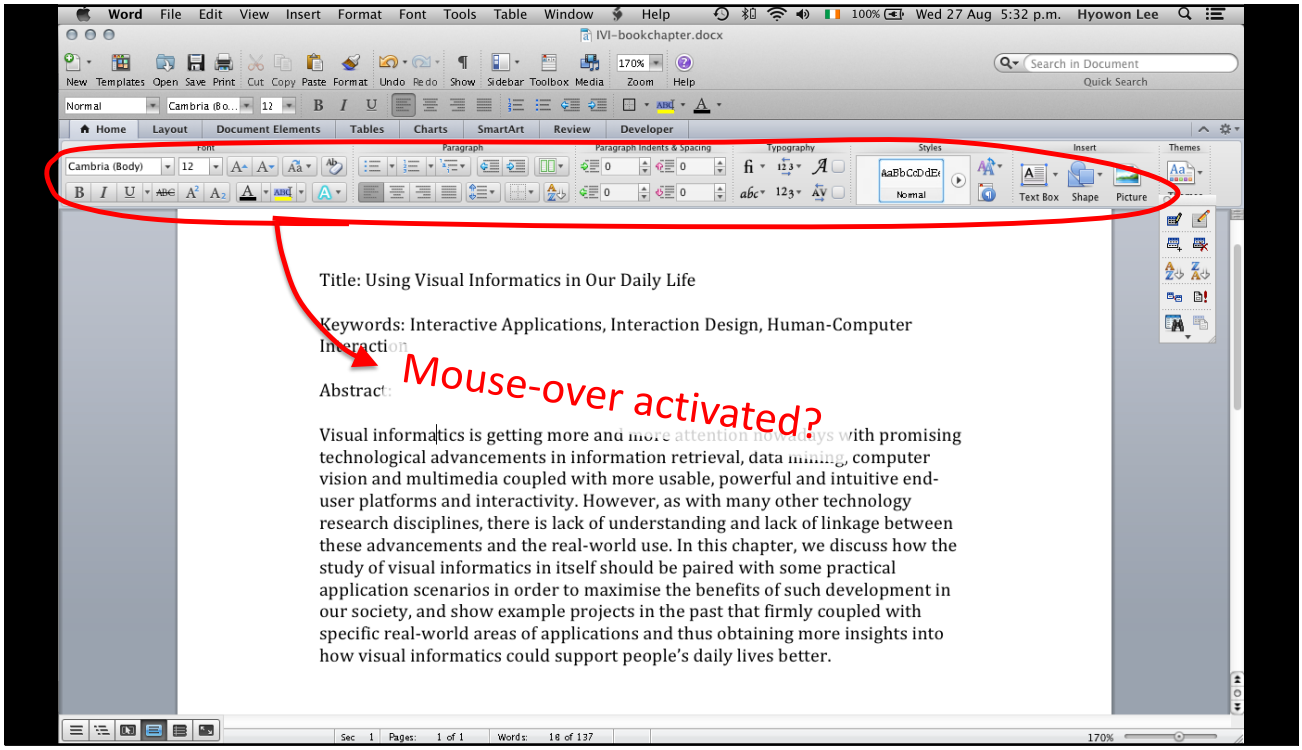
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Clicking VS. Mouse-Over

VS. User-Maintained

Mouse-Over is not available on our (multi-touch) phones or tablets.
=> Think about how mouse-over could be designed in for touch devices.
User-Maintained Interaction is where you click/press and keep holding it down in order to see the effect; when released, the effect ends.
=> This about how User-Maintained Interaction can be utilized more on our phone apps.

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20

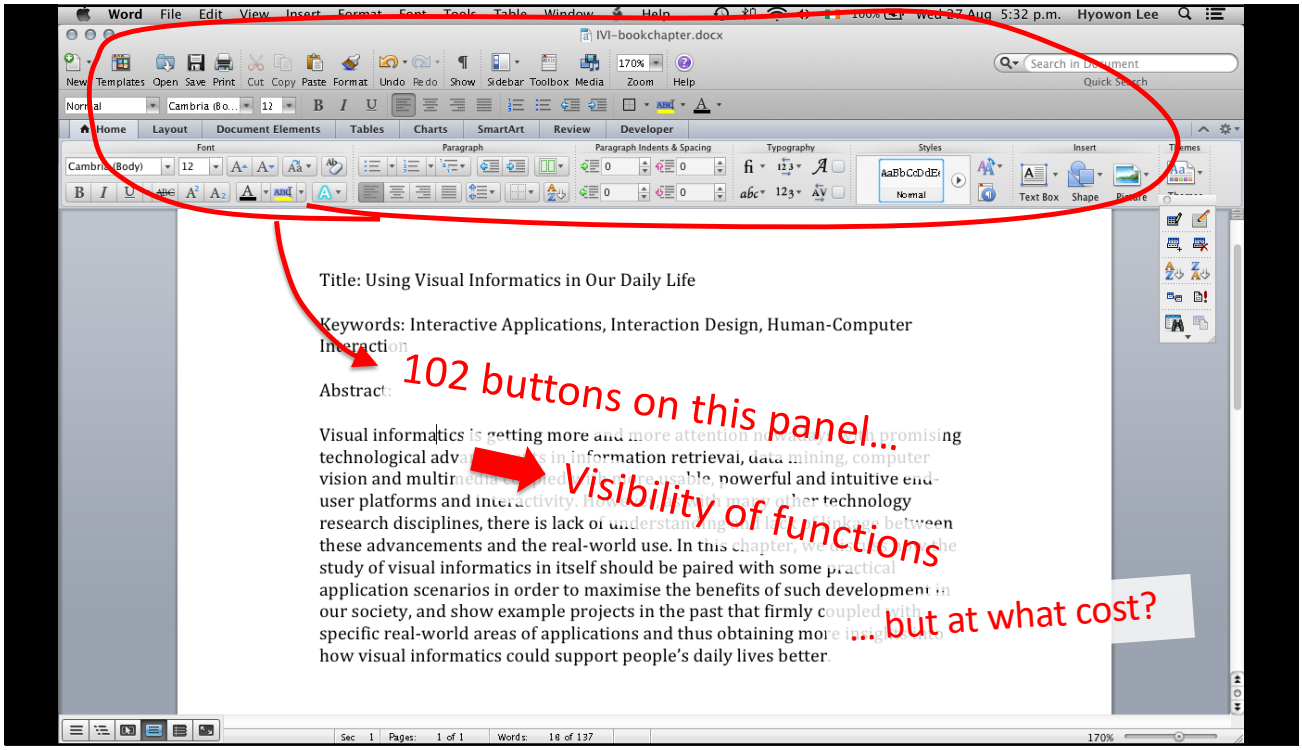
Guideline 5:
Reduce user's memory load

- Make the possible actions and/or status **visible**

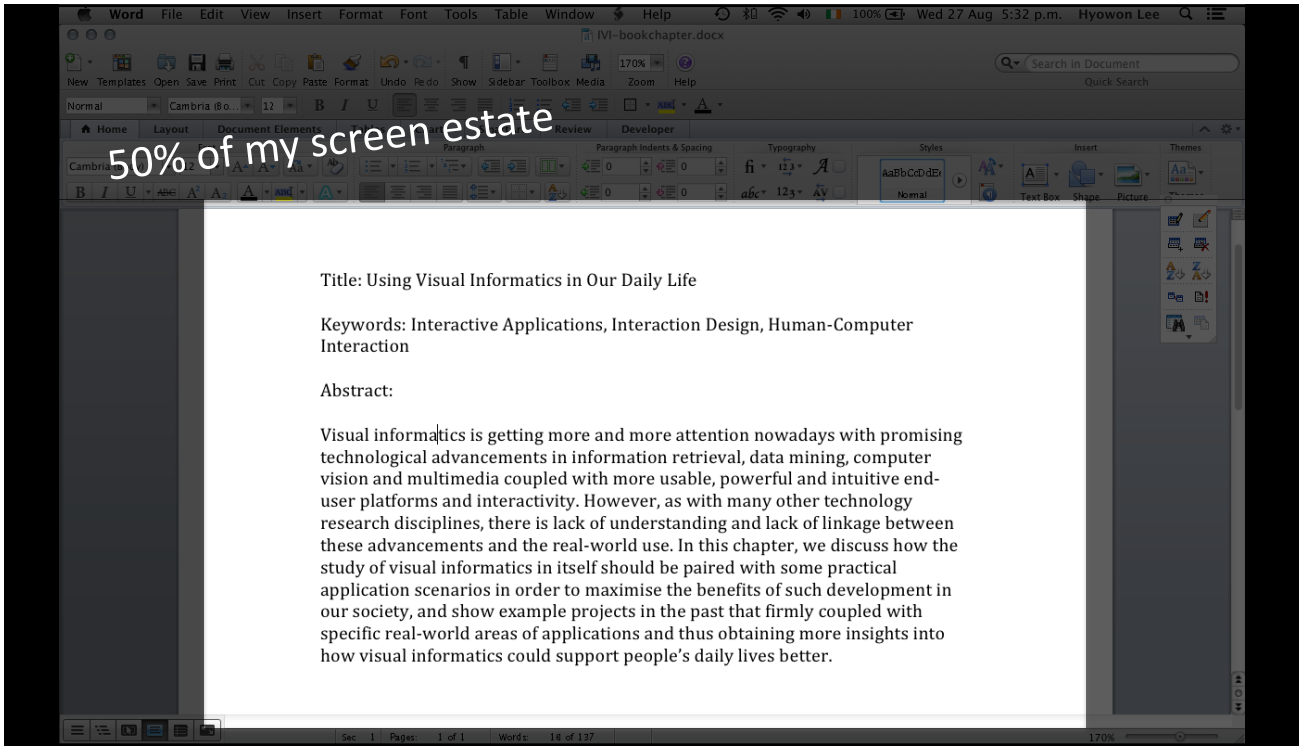


DCU

21



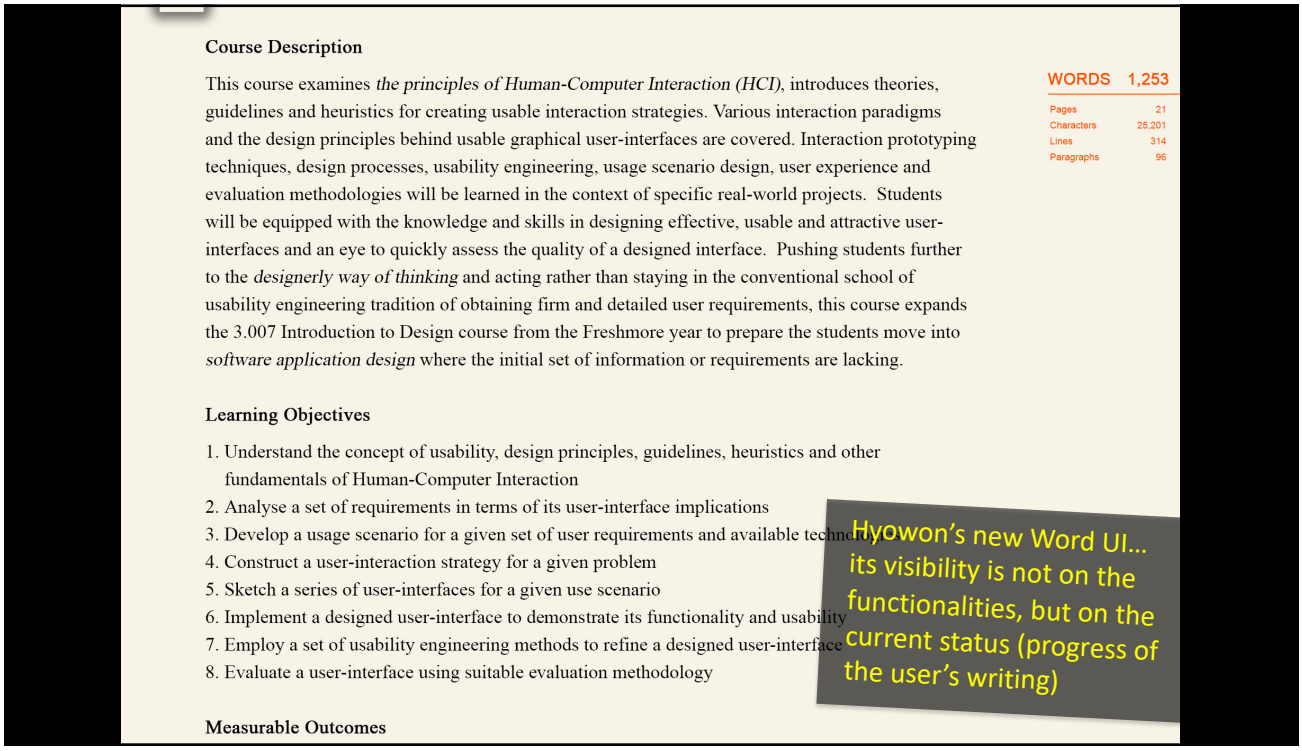
22


50% of my screen estate
Title: Using Visual Informatics in Our Daily Life
Keywords: Interactive Applications, Interaction Design, Human-Computer Interaction
Abstract:
Visual informatics is getting more and more attention nowadays with promising technological advancements in information retrieval, data mining, computer vision and multimedia coupled with more usable, powerful and intuitive end-user platforms and interactivity. However, as with many other technology research disciplines, there is lack of understanding and lack of linkage between these advancements and the real-world use. In this chapter, we discuss how the study of visual informatics in itself should be paired with some practical application scenarios in order to maximise the benefits of such development in our society, and show example projects in the past that firmly coupled with specific real-world areas of applications and thus obtaining more insights into how visual informatics could support people's daily lives better.

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ACTIVITY (10 min)

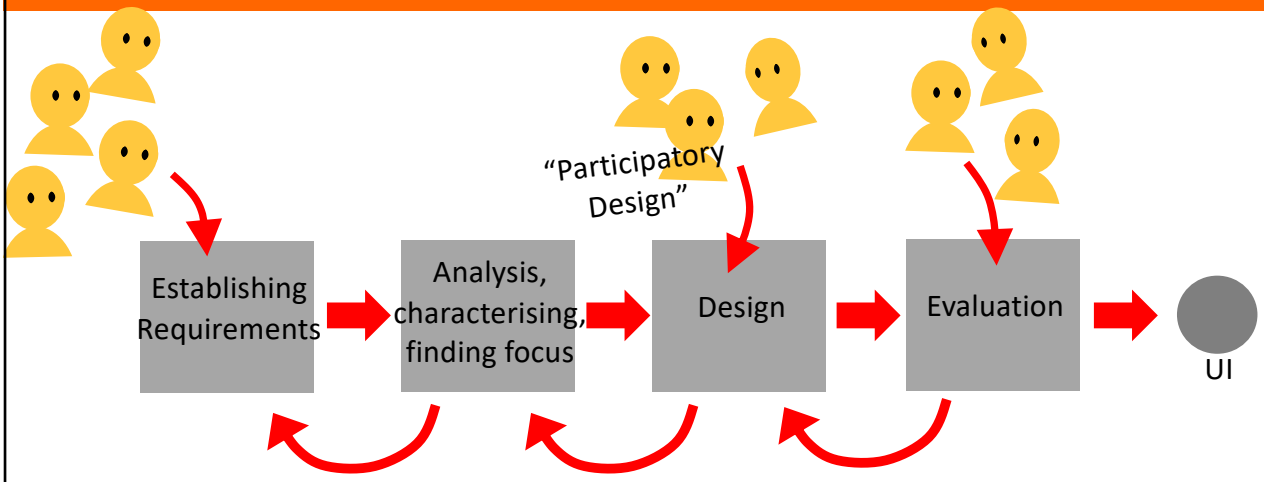
Re-design web browser UI with "Make the possible actions and/or status visible" in mind

- Structure of the page
- Hyperlinks
- User comments
- Page length
- :

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User-Centred Design

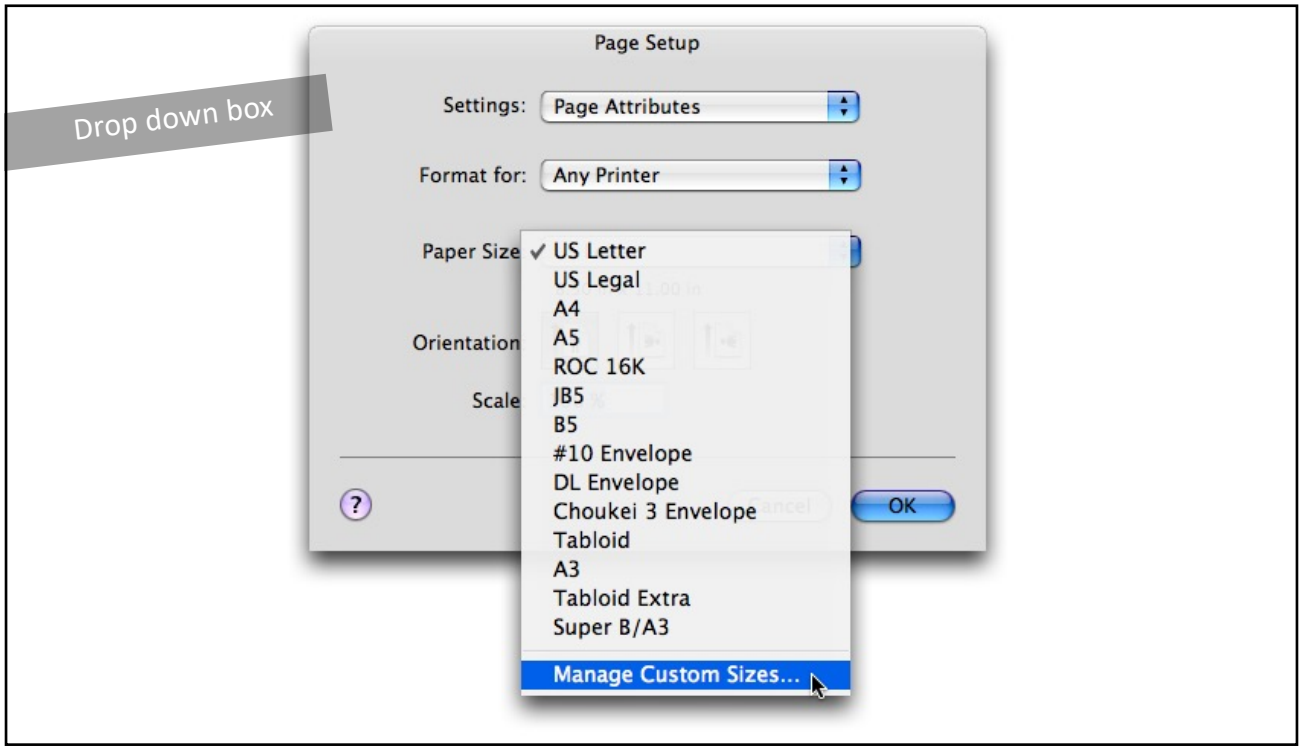


27

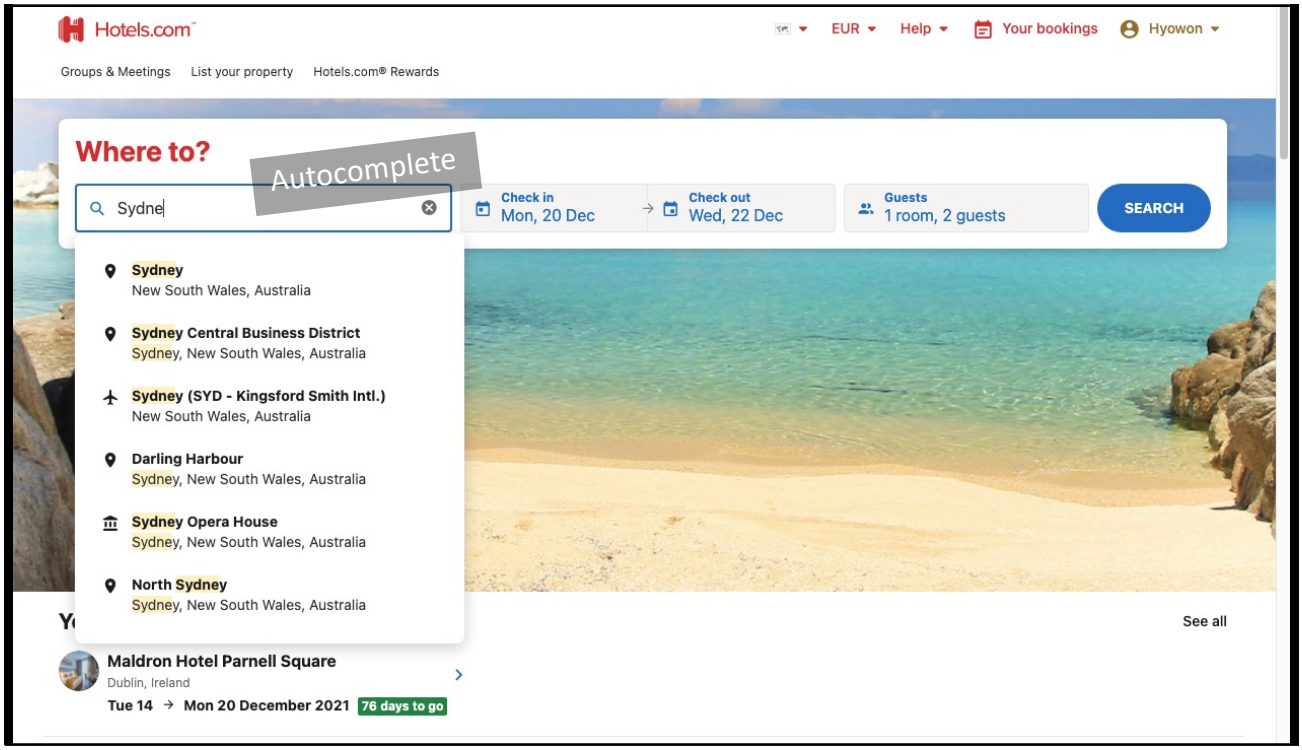
Guideline 5:
Reduce user's memory load

- Make the possible actions and/or status **visible**
- **Recognition** rather than **recall**

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Drop down box
Page Setup
Settings: Page Attributes
Format for: Any Printer
Paper Size
Orientation
Scale
US Letter
US Legal
A4
A5
ROC 16K
JB5
B5
#10 Envelope
DL Envelope
Choukei 3 Envelope
Tabloid
A3
Tabloid Extra
Super B/A3
Manage Custom Sizes...
OK

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Hotels.com
EUR
Help
Your bookings
Hyowon
Groups & Meetings
List your property
Hotels.com® Rewards
Where to?
Autocomplete
Sydne
Check in
Mon, 20 Dec
Check out
Wed, 22 Dec
Guests
1 room, 2 guests
SEARCH
Sydney
New South Wales, Australia
Sydney Central Business District
Sydney, New South Wales, Australia
Sydney (SYD - Kingsford Smith Intl.)
New South Wales, Australia
Darling Harbour
Sydney, New South Wales, Australia
Sydney Opera House
Sydney, New South Wales, Australia
North Sydney
Sydney, New South Wales, Australia
See all
Maldron Hotel Parnell Square
Dublin, Ireland
Tue 14 → Mon 20 December 2021
76 days to go

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Guideline 5:
Reduce user's memory load

- Make the possible actions and/or status **visible**
- **Recognition** rather than **recall**

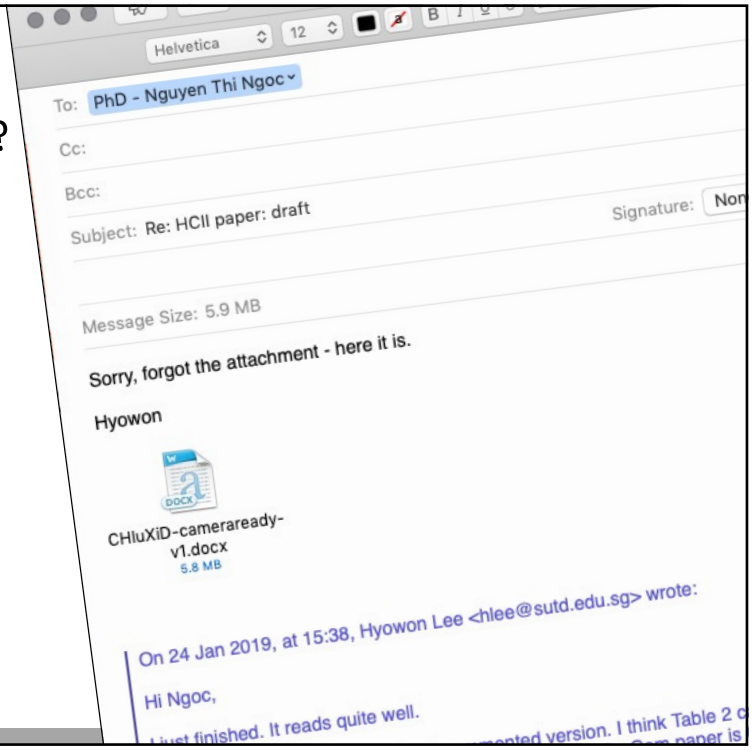
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DISCUSSION

Forgot email attachment?

e.g. ATM, photocopying



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